



Study of antibiotics sensitivity pattern and molecular characterization of *Staphylococcus aureus* isolated from human and animal pyogenic cases

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Abstract

Staphylococcus aureus has been described as the most common cause of human and animal diseases and has emerged as a superbug due to multidrug resistance. Considering these, a total of 175 samples were collected from pyogenic cases of humans (75) and animals (100), to establish the drug resistance pattern and also for molecular characterization of human and animal isolates. Thermonuclease (*nuc*) gene amplification was used to confirm all presumptive *S. aureus* isolates and then, antibiotic sensitivity and slide Coagulase tests were used for phenotypic characterization of isolates. Following that, all the isolates were subjected to PCR amplification to detect the existence of the Methicillin-resistant (*mecA*) and Coagulase (*coa*) genes. Lastly, typing was done using the Randomly Amplified Polymorphic DNA-PCR. The overall prevalence of *S. aureus* in human and animal samples was found to be 39.4%. Drug sensitivity revealed the highest resistance against the β -lactam antibiotics such as ampicillin (94.8%) and penicillin (90.6%), followed by cephalosporin (cefixime—67.7%) and quinolone (ciprofloxacin—52.1%) group of drugs. The drug sensitivity was the highest against antibiotics like chloramphenicol (95%) followed by gentamicin (90%). Among the 69 *S. aureus* isolates, the overall presence of MRSA was 40.5% (27.5% and 50% in human and animal isolates, respectively). Total 33 isolates exhibited *coa* genes amplification of more than one amplicons and variable in size of 250, 450, 800, and 1100 bp. The RAPD typing revealed amplification of five and six different band patterns in humans and animals, respectively, with two common patterns suggesting a common phylogenetic profile.

Keywords *Staphylococcus aureus* · Antibiotic sensitivity · *mecA* gene · *Coa* gene · RAPD-PCR

Introduction

Staphylococcus aureus has become the greatest global concern. According to Franklin and Lowy (1998), *S. aureus* has become an utmost problem due to its capacity to cause life-threatening infections, its inherent virulence, and its ability

to adjust to various ecological conditions. In recent times, *S. aureus* has gained attention due to its ability to gain resistance against commonly used antibiotics. The *S. aureus* strain is considered as drug-resistant upon acquisition of the *mecA* gene, this gene encodes the penicillin-binding protein 2a (PBP2a), and its incorporation within a large chromosomal element, called Staphylococcal Cassette Chromosome *mec* (SCC *mec*). Hartman and Tomasz (1984) showed that the *mecA* gene is present in the resistant strains but absent in the susceptible ones. Robinson and Enright (2004) observed that “the methicillin-sensitive strains of *S. aureus* acquired the SCC *mec* element probably from Coagulase-Negative Staphylococcal strains and became methicillin-resistant”. Moreover, “the *nuc* gene amplification by PCR is considered a standard technique for the identification of *S. aureus*. The *nuc* gene in *S. aureus* encodes the thermonuclease enzymes and amplification of the *nuc* gene is potential for rapid diagnosis of *S. aureus* infection” (Kateete et al. 2010).

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Staphylococcus aureus also produces extracellular proteins; one of them is Coagulase, which can bind with prothrombin and fibrinogen. “The Coagulase and prothrombin complex binds with fibrinogen and convert it to fibrin threads by a mechanism different from natural clotting” (Palma et al. 1999). The Coagulase is encoded by the *coa* gene which possesses a conserved and repeated polymorphic region (Reinoso et al. 2008).

To combat MRSA infections, quick and accurate typing of *S. aureus* strains is compulsory. Many typing methods have been described (Frénay et al. 1994; Kluytmans et al. 1995; Van Leeuwen et al. 2003); among these methods for genotyping of *S. aureus*, the Randomly Amplified Polymorphic DNA (RAPD) technique has been observed as a basic, effective, and quick method (Tambic et al. 1997). Based on these observations and increasing resistance to different antibiotics, the present study was designed to evaluate the antibiotics sensitivity patterns, to determine the correlation of antimicrobial resistance *mecA* gene with antibiotics sensitivity patterns and with virulence *coa* gene in *S. aureus* isolated from pyogenic cases of humans and animals. Finally, the phylogenetic correlation between human and animal MRSA isolates was evaluated by RAPD-PCR.

Materials and methods

Sample collection and identification of *S. aureus*

A total of 175 samples were collected, consisting of 75 pus samples from humans and 100 pus samples from animals (cattle, dogs, and, buffalos). The study was performed at the Department of Veterinary Microbiology and Immunology, Pandit Deen Dayal Upadhyaya Pashu Chikitsa Vigyan Vishwavidyalaya Evam Go-Anusandhan Sansthan (DUVASU), Mathura, Uttar Pradesh, India. All the collected samples were immediately shifted to Tryptone soya broth (Hi-Media, India) containing 6.5% of NaCl (Merck, India) and then incubated at 37 °C overnight. Then, obtained culture growths were subjected to isolation and identification of *S. aureus* based on cultural, morphological, and biochemical characteristics. These cultures were further subjected to Coagulase tests as per standard procedures (Quinn et al. 2002).

The extraction of genomic DNA from all the isolates was done by the Phenol–Chloroform Method suggested by Sambrook and Russell (2021) with slight modifications. All the presumptive *S. aureus* isolates were subjected to PCR-based amplification of the 280 bp species-specific *nuc* gene using custom synthesized previously described primers (Brakstad et al. 1992). All the obtained amplicons were analyzed under a UV transilluminator (Spectroline, Westbury, NY) after

electrophoresis in a 1.5% agarose gel containing ethidium bromide and standard molecular weight markers.

Antibiotic sensitivity testing

The disc diffusion method was used to perform the antibiotic sensitivity test as recommended by the Clinical and Laboratory Standards Institute (CLSI 2011). Different groups of antimicrobial drugs were used containing a designated amount of drug (HiMedia, India), including vancomycin (30 µg), methicillin (10 µg), amoxicillin (10 µg), ticarcillin (75 µg), penicillin G (10U), erythromycin (15 µg), clindamycin (2 µg), amikacin (10 µg), streptomycin (10 µg), gentamicin (10 µg), cefazolin (30 µg), chloramphenicol (30 µg), cefoxitin (30 µg), tetracycline (30 µg), cefixime (5 µg), and ciprofloxacin (5 µg). Using an antibiotic zone scale (Himedia, India), the zone of inhibition was measured in millimeters after 24 h of incubation.

Determination of methicillin-resistant *mecA* and Coagulase-producing *coa* genes

All the confirmed *S. aureus* isolates were subjected to amplification of antimicrobial resistance *mecA* (Predari et al. 1991) and Coagulase-producing *coa* gene (Montesinos et al. 2002) on genomic DNA. The PCR reactions were carried out in a total volume of 25 µl made up of the 2 µl template DNA, 6.5 µl nuclease-free water, 2 µl each set of primers, and 12.5 µl HotStart PCR Master Mix (Takara Bio Inc., Japan). The PCR amplification was performed using previously reported procedures for *mecA* genes (Predari et al. 1991), and *coa* genes (Montesinos et al. 2002).

RAPD typing

RAPD typing was done using AP4 primer 5'-TCACGC TGCA-3'. PCR reactions were carried out in an absolute volume of 50 µl comprised of the 2 µl template genomic DNA, 4 µl AP4 primer, 25 µl 2X HotStart PCR Master Mix (Takara Bio Inc., Tokyo, Japan), and 19 µl nuclease-free water to make up the volume. All the obtained amplicons were analyzed under a UV transilluminator (Spectroline, Westbury, NY) after electrophoresis in a 1.5% agarose gel containing ethidium bromide and standard molecular weight markers. For negative controls, AP4 PCR reactions without DNA were used (Barbier et al. 1996).

Results and discussion

MRSA was first isolated from cows with mastitis in 1972 (Devriese et al. 1972), followed by isolation of MRSA of human origin from dairy cows was performed by Devriese

and Homme (1975). Since the first report of MRSA, the presence of MRSA has been reported almost from all the domestic species. In this study, a total of 175 samples were collected from human and animal pyogenic cases. The study revealed the 39.4% ($n=69$) overall prevalence of *S. aureus* in human and animal samples. These included 38.6% (J1–J29) and 40% (J30–J69) prevalence in human and animal pyogenic cases, respectively (Fig. 1).

All the 69 *S. aureus* isolates were tested for antimicrobial susceptibility and the resistance was highest against the β -lactam group of antibiotics such as ampicillin (94.8%), penicillin (90.6%), methicillin (81.3%), and ticarcillin

(70.8%) (Fig. 2, Table 1). In contrast to this study, 100 samples obtained from diverse pyogenic conditions of cattle, buffaloes, and canines showed 95% resistance to amoxicillin and 82.5% resistance to penicillin (Yadav et al. 2018a, b). Following the β -lactam, resistance was observed against cephalosporin drugs like cefixime (67.7%) and quinolone groups of drugs such as ciprofloxacin (52.1%) (Table). In 2012, Jayatilake and Bandara (2012) conducted a study in a tertiary care hospital of Sri Lanka and reported the same resistance to ciprofloxacin (54%) among 125 MRSA, isolated from various clinical samples. Irrespective of a group of drug resistance was higher in human isolates in comparison to animal isolates. The ciprofloxacin is commonly used in humans against *S. aureus* was found to be 78.6% resistant in humans and 15.0% in animals. A study has shown that after 3 months of ciprofloxacin use, the resistance rate was increased from none to 79% over 1 year (Blumberg et al. 1991). However, other antibiotics were found to be sensitive. The highest sensitivity was observed against drugs like chloramphenicol (95%) followed by gentamicin (90%), cefoxitin (86.5%), streptomycin (86.5%), and tetracycline (83.9%). This study is in agreement with another report that reported the highest sensitivity to chloramphenicol (85.2%) (Iliya et al. 2020). In one study, 83% and 81% sensitivity was obtained against gentamicin and tetracycline, respectively (Tiware et al. 2011). The human isolates revealed the highest sensitivity to quinolones like cefoxitin (85.7%) followed by aminoglycosides like streptomycin (83.9%). The animal isolates revealed the highest sensitivity to aminoglycosides like streptomycin (90%) followed by cefoxitin

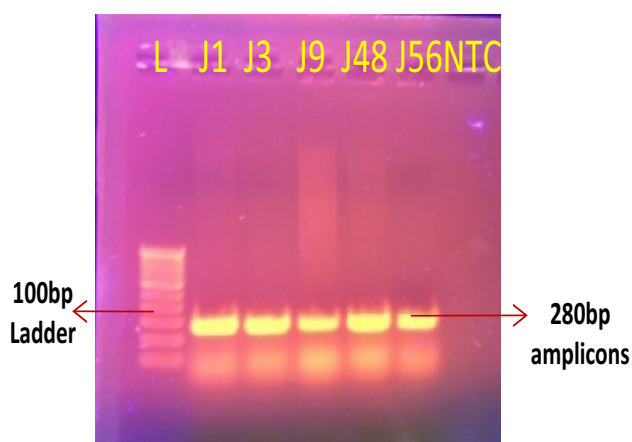


Fig. 1 Amplification of the *S. aureus*-specific nuc gene yielded a 280-bp product

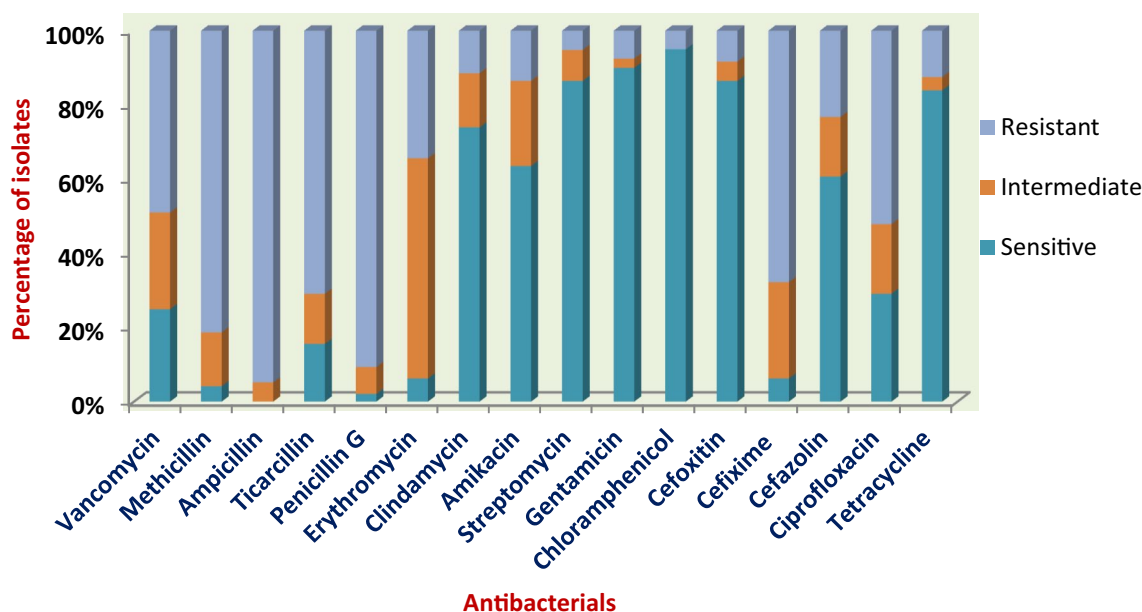


Fig. 2 Overall drug sensitivity of *S. aureus* isolates. The sensitivity was highest against chloramphenicol (95%) and resistance was highest against ampicillin (94.8%)

Table 1 Drug sensitivity pattern (in %) of *S. aureus* isolates against different groups of antibiotics

S. no.	Antibiotic name	Sensitivity			Intermediate			Resistance		
		H	A	Total	H	A	Total	H	A	Total
1	Vancomycin (30 µg)	1.8	57.5	25.0	14.3	42.5	26.0	83.9	0.0	49.0
2	Methicillin (10 µg)	0.0	10.0	4.2	0.0	35.0	14.6	100.0	55.0	81.3
	Ampicillin (10 µg)	0.0	0.0	0.0	3.6	7.5	5.2	96.4	92.5	94.8
	Ticarcillin (75 µg)	3.6	32.5	15.6	5.4	25.0	13.5	91.1	42.5	70.8
	Penicillin G (10 units)	0.0	5.0	2.1	1.8	15.0	7.3	98.2	80.0	90.6
3	Erythromycin (15 µg)	1.8	12.5	6.3	51.8	70.0	59.4	46.4	17.5	34.4
4	Clindamycin (2 µg))	67.9	82.5	74.0	0.0	15.0	14.6	17.9	2.5	11.5
5	Amikacin (10 µg)	53.6	77.5	63.5	0.0	20	22.9	21.4	2.5	13.5
	Streptomycin (10 µg)	83.9	90.0	86.5	10.7	5.0	8.3	5.4	5.0	5.2
	Gentamicin (30 µg)	10.7	75.0	90.0	0.0	2.5	2.5	0.0	7.5	7.5
6	Chloramphenicol (5 µg)	3.5	85.0	95.0	0.0	0.0	0.0	0.0	5.0	5.0
7	Cefoxitin (30 µg)	85.7	87.5	86.5	5.4	5.0	5.2	8.9	7.5	8.3
	Cefixime (10 µg)	3.6	10.0	6.3	1.8	60.0	26.0	94.6	30.0	67.7
	Cefazolin (30 µg)	53.5	10.0	60.7	16.1	0.0	16.1	23.2	0.0	23.2
8	Ciprofloxacin (30 µg)	12.5	52.5	29.2	8.9	32.5	18.8	78.6	15.0	52.1
9	Tetracycline (10 µg)	71.4	17.5	83.9	3.6	0.0	3.6	12.5	0.0	12.5

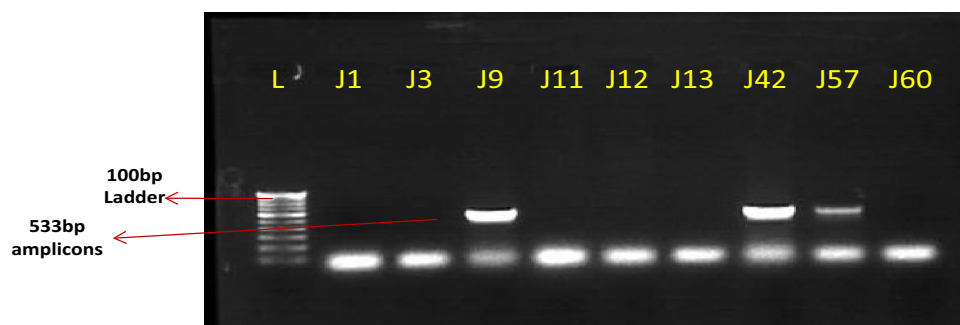
H—human *S. aureus* isolates, A—animal *S. aureus* isolates

(87.5%), chloramphenicol (85%), and clindamycin (74%) (Fig. 2, Table 1).

In this study, out of 69 *S. aureus* isolates, 28 (40.5%) isolates showed amplification of the *mecA* gene (Fig. 3). These included 8 (27.5%) isolates from human pyogenic cases and 20 (50%) from different animal species. This study in concurrence with another study of Nepal reported 26.14% of the prevalence of MRSA was procured from various human samples of a tertiary-care hospital (Kumari et al. 2008). In the case of humans, out of 29 *S. aureus* isolates, only 8 showed antimicrobial resistance *mecA* gene amplification, however, out of 40 animal *S. aureus* isolates, 18 showed *mecA* gene amplification, while all human and animal isolates were resistant to methicillin antibiotic. In another study out of 40 animal isolates, 23 isolates revealed amplification of the *mecA* gene (Yadav et al. 2018a, b).

The detection of Coagulase can be made by slide/tube Coagulase test, however, both human and sheep plasmas had very low specificity (11% and 7%) so may lead to

misleading results (Kateete et al. 2010). Hence, it cannot be used as a single test to confirm *S. aureus*. Similarly, in the present study, out of 69 isolates, 33 (47.8%) isolates revealed a positive slide Coagulase test (Fig. 4). These included 24 (34.7%) human and 9 (13%) animal isolates. These isolates were also confirmed for the presence of pathogenicity-related *coa* genes on genomic DNA. Total 33 (47.8%) isolates showed amplicons of variable size and number for *coa* genes amplicons irrespective of positive or negative slide Coagulase test results. The *coa* genes positive isolates included 24 (34.7%) isolates from human pyogenic cases and 9 (13%) from animal isolates. Size of amplicons showed polymorphism with amplification of more than one amplicons product ranging from 250 to 1100 bp. Likewise in one study, Izadpanah and Asadpour (2018) found that two amplicons of size 680 bp and 750 bp were obtained from 21 (out of 30) clinical isolates of *S. aureus*. In another study, Abdulghany and Khairy (2014) found that “out of 58 MRSA isolates, 15 *coa* types were

Fig. 3 Amplification of methicillin-resistant *mecA* genes from *S. aureus* isolates resulted in a 533 bp product

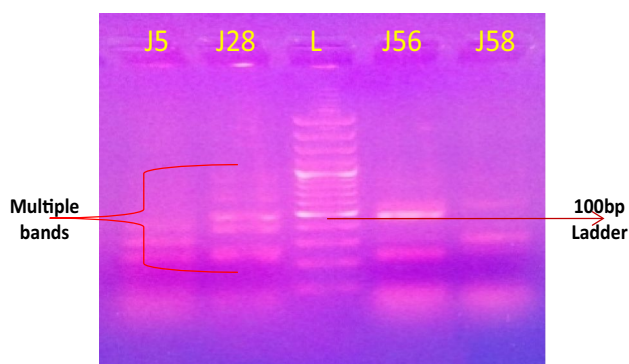


Fig. 4 Amplification of the *coa* gene yielded multiple amplicons with sizes ranging from 250 to 450, 800, and 1100 bp

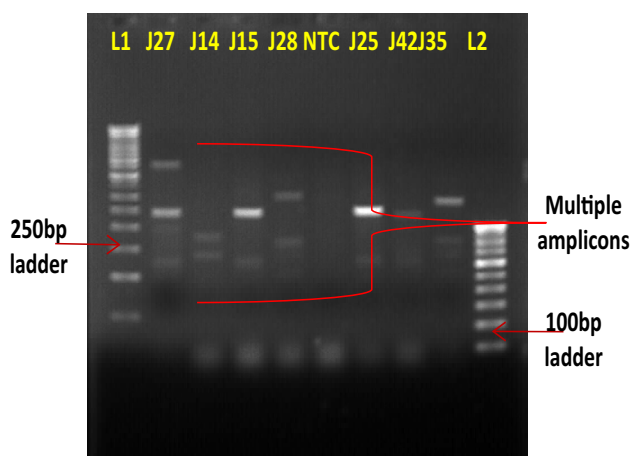


Fig. 5 Typing of *S. aureus* isolates using RAPD-AP4 PCR revealed 2–5 amplicons

classified, and the amplification products showed multiple bands (1, 2, 3, 4, 5, and 8 bands)".

The RAPD-AP4 PCR revealed that six out of eight human MRSA isolates had amplification of 2–5 bands with five different band patterns (Fig. 5). A previous study of Polyclinic hospital in Italy, also suggested the five distinct amplicons were found in human MRSA isolates (Corrente et al. 2005). In the case of animals, out of 20 MRSA, 18 isolates revealed amplification of 2–5 bands with six different band patterns (Fig. 5). Out of these six patterns, two were similar to patterns of human isolates suggesting a common phylogenetic profile of MRSA isolates in humans and animals. In contrast to this study, one study reported RAPD patterns of six animal isolates were found to be similar to some human isolates (Lee 2003). The common patterns obtained in humans and animals also suggest that MRSA might have transferred from animal to human. Several studies have suggested that "the transfer of *S. aureus* between humans and cattle is possible" (Fox et al. 1991; Roberson et al. 1994; Zadoks et al. 2000).

Conclusion

In contrast to human samples, animal samples had a higher prevalence of *S. aureus*. Animal isolates were more susceptible to the antibiotics used in the study than human isolates. Resistance was higher in human isolates than in animal isolates, regardless of the drug type. Antibiotics often used to treat infections caused by *S. aureus* showed more resistance, whereas antibiotics that were no longer in use showed more sensitivity. As a result, other antibiotics must be used instead of widely prescribed antibiotics to treat *S. aureus* infections. It is also concluded that the presence and absence of *coa* genes were not related to the presence of methicillin-resistant *mecA* genes or the drug resistance pattern of isolates. In RAPD-PCR typing, two-band patterns were similar in human and animal MRSA isolates, suggesting a common phylogenetic profile or transfer from animal to human or vice-versa.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval No ethical clearance was required as samples were collected from the clinical cases of both humans and animals.

References

- Abdulghany HM, Khairy RM (2014) The frequency of methicillin-resistant *Staphylococcus aureus* and coagulase gene polymorphism in Egypt. Int J Bacteriol 2014:1–6. <https://doi.org/10.1155/2014/680983>
- Barbier N, Saulnier P, Chachaty E et al (1996) Random amplified polymorphic DNA typing versus pulsed-field gel electrophoresis for epidemiological typing of vancomycin-resistant enterococci. J Clin Microbiol 34:1096–1099. <https://doi.org/10.1128/jcm.34.5.1096-1099.1996>
- Blumberg HM, Rimland D, Carroll DJ et al (1991) Rapid development of ciprofloxacin resistance in methicillin-susceptible and -resistant *Staphylococcus aureus*. J Infect Dis 163:1279–1285. <https://doi.org/10.1093/infdis/163.6.1279>
- Brakstad OG, Aasbakk K, Maeland JA (1992) Detection of *Staphylococcus aureus* by polymerase chain reaction amplification of the *nuc* gene. J Clin Microbiol 30:1654–1660. <https://doi.org/10.1128/jcm.30.7.1654-1660.1992>

- Clinical and Laboratory Standards Institute (CLSI) (2011) Performance standards for antimicrobial Susceptibility Testing. An informational supplement for global application developed through the Clinical and Laboratory Standards Institute
- Corrente M, Monno R, Totaro M et al (2005) Characterization of methicillin-resistant *Staphylococcus aureus* (MRSA) isolated at the Policlinico Hospital of Bari (Italy). *New Microbiol* 28:57–65
- Devriese LA, Hommez J (1975) Epidemiology of methicillin-resistant *Staphylococcus aureus* in dairy herds. *Res Vet Sci* 19:23–27
- Devriese LA, Van Damme LR, Fameree L (1972) Methicillin (Cloxacillin)-resistant *Staphylococcus aureus* strains isolated from bovine mastitis cases. *Zentralblatt Für Veterinärmedizin R B* 19:598–605. <https://doi.org/10.1111/j.1439-0450.1972.tb00439.x>
- Fox LK, Gershman M, Hancock DD, Hutton CT (1991) Fomites and reservoirs of *Staphylococcus aureus* causing intramammary infections as determined by phage typing: the effect of milking time hygiene practices. *Cornell Vet* 81:183–193
- Franklin D, Lowy MD (1998) *Staphylococcus aureus* infections. *N Engl J Med* 339:520–532. <https://doi.org/10.1056/NEJM199808203390806>
- Frénay H, Theelen J, Schouls L et al (1994) Discrimination of epidemic and nonepidemic. *J Clin Microbiol* 32:846–847
- Hartman BJ, Tomasz A (1984) Low-affinity penicillin-binding protein associated with β -lactam resistance in *Staphylococcus aureus*. *J Bacteriol* 158:513–516. <https://doi.org/10.1128/jb.158.2.513-516.1984>
- Iliya S, Mwangi J, Maathai R, Muriuki M (2020) Phenotypic analysis and antibiotic susceptibility of methicillin-resistant *Staphylococcus aureus* in Kiambu County, Kenya. *J Infect Dev Ctries* 14(6):597–605. <https://doi.org/10.3855/jidc.12174>
- Izadpanah MR, Asadpour L (2018) Investigation of *coa* gene polymorphism in clinical isolates of *Staphylococcus aureus* in North of Iran. *J Cell Mol Res* 10:27–31. <https://doi.org/10.22067/jcmr.v10i1.52730>
- Jayatilleke K, Bandara P (2012) Antibiotic sensitivity pattern of *Staphylococcus aureus* in a tertiary care hospital of Sri Lanka. *Sri Lankan J Infect Dis* 2:13. <https://doi.org/10.4038/sljd.v2i2.4162>
- Kateete DP, Kimani CN, Katabazi FA et al (2010) Identification of *Staphylococcus aureus*: DNase and Mannitol salt agar improve the efficiency of the tube Coagulase test. *Ann Clin Microbiol Antimicrob* 9:1–7. <https://doi.org/10.1186/1476-0711-9-23>
- Kluytmans J, Van Leeuwen W, Goessens W et al (1995) Food-initiated outbreak of methicillin-resistant *Staphylococcus aureus* analyzed by pheno- and genotyping. *J Clin Microbiol* 33:1121–1128. <https://doi.org/10.1128/jcm.33.5.1121-1128.1995>
- Kumari N, Mohapatra TM, Singh YI (2008) Prevalence of Methicillin-resistant *Staphylococcus aureus* (MRSA) in a tertiary-care hospital in eastern Nepal. *J Nepal Med Assoc* 47:53–56. <https://doi.org/10.31729/jnma.309>
- Lee JH (2003) Methicillin (oxacillin)-resistant *Staphylococcus aureus* strains isolated from major food animals and their potential transmission to humans. *Appl Environ Microbiol* 69:6489–6494. <https://doi.org/10.1128/AEM.69.11.6489-6494.2003>
- Montesinos I, Salido E, Delgado T et al (2002) Epidemiologic genotyping of methicillin-resistant *Staphylococcus aureus* by pulsed-field gel electrophoresis at a University Hospital and comparison with antibiotyping and protein A and Coagulase gene polymorphisms. *J Clin Microbiol* 40:2119–2125. <https://doi.org/10.1128/JCM.40.6.2119-2125.2002>
- Palma M, Hagggar A, Flock JI (1999) Adherence of *Staphylococcus aureus* is enhanced by an endogenous secreted protein with broad binding activity. *J Bacteriol* 181:2840–2845. <https://doi.org/10.1128/jb.181.9.2840-2845.1999>
- Predari SC, Ligozzi M, Fontana R (1991) Genotypic identification of methicillin-resistant coagulase-negative staphylococci by polymerase chain reaction. *Antimicrob Agents Chemother* 35:2568–2573. <https://doi.org/10.1128/AAC.35.12.2568>
- Quinn PJ, Markey BK, Carter ME et al (2002) Veterinary microbiology and microbial disease. Blackwell Science, Oxford
- Reinosa EB, El-Sayed A, Lämmle C et al (2008) Genotyping of *Staphylococcus aureus* isolated from humans, bovine subclinical mastitis, and food samples in Argentina. *Microbiol Res* 163:314–322. <https://doi.org/10.1016/j.micres.2006.05.013>
- Roberson JR, Fox LK, Hancock DD et al (1994) Ecology of *Staphylococcus aureus* isolated from various sites on dairy farms. *J Dairy Sci* 77:3354–3364. [https://doi.org/10.3168/jds.S0022-0302\(94\)77277-5](https://doi.org/10.3168/jds.S0022-0302(94)77277-5)
- Robinson DA, Enright MC (2004) Evolution of *Staphylococcus aureus* by large chromosomal replacements. *J Bacteriol* 186:1060–1064. <https://doi.org/10.1128/JB.186.4.1060-1064.2004>
- Sambrook J, Russell DW (2021) Molecular Cloning: a Laboratory Manual 1:2021
- Tambic A, Power EGM, Talsania H et al (1997) Analysis of an outbreak of non-phage-typeable methicillin-resistant *Staphylococcus aureus* by using a Randomly Amplified Polymorphic DNA assay. *J Clin Microbiol* 35:3092–3097. <https://doi.org/10.1128/jcm.35.12.3092-3097.1997>
- Tiwari S, Sahu M, Rautaraya B et al (2011) Prevalence of methicillin-resistant *Staphylococcus aureus* and its antibiotic susceptibility pattern in a tertiary care hospital. *J Indian Med Assoc* 109:800–801
- Van Leeuwen WB, Jay C, Snijders S et al (2003) Multilocus sequence typing of *Staphylococcus aureus* with DNA array technology. *J Clin Microbiol* 41:3323–3326. <https://doi.org/10.1128/JCM.41.7.3323-3326.2003>
- Yadav R, Kumar A, Singh VK et al (2018a) Molecular determination of Methicillin-resistance *mecA* and virulence *coa* genes in *Staphylococcus aureus* from pyogenic clinical cases of companion animals in India. *Turkish J Vet Anim Sci* 42:371–375. <https://doi.org/10.3906/vet-1705-13>
- Yadav R, Kumar A, Singh VK et al (2018b) Prevalence and antibiotyping of *Staphylococcus aureus* and methicillin-resistant *S. aureus* (MRSA) in domestic animals in India. *J Glob Antimicrob Resist* 15:222–225. <https://doi.org/10.1016/j.jgar.2018.08.001>
- Zadoks R, Van Leeuwen W, Barkema H et al (2000) Application of pulsed-field gel electrophoresis and binary typing as tools in veterinary clinical microbiology and molecular epidemiologic analysis of bovine and human *Staphylococcus aureus* isolates. *J Clin Microbiol* 38:1931–1939. <https://doi.org/10.1128/jcm.38.5.1931-1939.2000>

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